

HDOC 38

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with options 1 to 4.
3. Read the choice and clear the input buffer.
4. Case 1 (Count Types): Read a string. Iterate through the string. Check if the character is a space, a digit ('0'-'9'), an alphabet ('a'-'z' or 'A'-'Z'), or a special character. Increment respective counters. Print the space, digit, and special character counts.
5. Case 2 (Count Frequency): Read a string and a target character. Iterate through the string. If the current character matches the target character exactly, increment a counter. Print the total count.
6. Case 3 (Toggle Case): Read a string. Iterate through the string. If a character is lowercase ('a'-'z'), subtract 32 from its ASCII value to make it uppercase. If it is uppercase ('A'-'Z'), add 32 to make it lowercase. Leave other characters unchanged. Print the modified string.
7. Case 4 (Exit): Terminate program execution.
8. Stop

2. Test-Case Table

TC ID	Scenario	Inputs	Expected Output
TC-01	Count Types	Choice: 1, String: "aB 12 @#"	Spaces: 1, Digits: 2, Special Characters: 2
TC-02	Count Frequency	Choice: 2, String: "hello", Char: "l"	Frequency of 'l' is 2.
TC-03	Toggle Case	Choice: 3, String: "Hello 123 World!"	Toggled case string: hELLO 123 wORLD!
TC-04	Exit	Choice: 4	Exiting program...

3. C Program Source Code

```
#include <stdio.h> void

countTypes(void);
void countFrequency(void); void
toggleCase(void);
void getString(char str[], int size);

int main() {
    int choice; do {
        printf("\n--- String Operations Menu 3 ---\n");
        printf("1. Count spaces, digits, and special characters\n"); printf("2. Count frequency of a
        character\n");
        printf("3. Toggle the case of each character\n"); printf("4. Exit\n");
        printf("Enter your choice: "); scanf("%d",
        &choice);

        while (getchar() != '\n');

        switch (choice) { case 1:
            countTypes();
            break;
            case 2:
                countFrequency(); break;
            case 3:
                toggleCase(); break;
            case 4:
                printf("Exiting program...\n"); break;
            default:
                printf("Error: Invalid choice.\n");
            }
        } while (choice != 4);

        return 0;
    }

void getString(char str[], int size) { printf("Enter a string:
    "); fgets(str, size, stdin);
    int i = 0;
    while (str[i] != '\0') {
        if (str[i] == '\n') {
            str[i] = '\0'; break;
        } i++;
    }
}
```

```

void countTypes(void) { char str[100];
    getString(str,100);
    int spaces = 0, digits = 0, special = 0; int i = 0;
    while (str[i] != '\0') {
        if (str[i] == ' ') { spaces++;
        } else if (str[i] >= '0' && str[i] <= '9') { digits++;
        } else if ((str[i] >= 'a' && str[i] <= 'z') || (str[i] >= 'A' && str[i] <= 'Z')) {

            } else {
                special++;
            } i++;
        }
    printf("Spaces: %d\nDigits: %d\nSpecial Characters: %d\n", spaces, digits, special);
}

```

```

void countFrequency(void) { char str[100];
    char ch;
    int count = 0, i = 0;

    getString(str,100);
    printf("Enter the character to find its frequency: "); scanf("%c", &ch);
    while (getchar() != '\n');

    while (str[i] != '\0') { if (str[i] == ch) {
        count++;
    } i++;
    }
    printf("Frequency of '%c' is %d.\n", ch, count);
}

```

```

void toggleCase(void) { char str[100];
    getString(str,100); int i = 0;
    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z') { str[i] = str[i] - 32;
        } else if (str[i] >= 'A' && str[i] <= 'Z') { str[i] = str[i] + 32;
        } i++;
    }
    printf("Toggled case string: %s\n", str);
}

```

4. Compilation and Execution

```
(kali@kali)~[/liak]  
$ cc hdoc_38.c
```

```
(kali@kali)~[/liak]  
$ ./a.out
```

```
— String Operations Menu 3 —  
1. Count spaces, digits, and special characters  
2. Count frequency of a character  
3. Toggle the case of each character  
4. Exit  
Enter your choice: 1  
Enter a string: Keshavs 1st github repo 2  
Spaces: 4  
Digits: 2  
Special Characters: 0
```

```
— String Operations Menu 3 —  
1. Count spaces, digits, and special characters  
2. Count frequency of a character  
3. Toggle the case of each character  
4. Exit  
Enter your choice: 2  
Enter a string: keshav hdoc 38 repo commit  
Enter the character to find its frequency: m  
Frequency of 'm' is 2.
```

```
— String Operations Menu 3 —  
1. Count spaces, digits, and special characters  
2. Count frequency of a character  
3. Toggle the case of each character  
4. Exit  
Enter your choice: 3  
Enter a string: hi i am program  
Toggled case string: HI I AM PROGRAM
```

```
— String Operations Menu 3 —  
1. Count spaces, digits, and special characters  
2. Count frequency of a character  
3. Toggle the case of each character  
4. Exit  
Enter your choice: 4  
Exiting program...
```